AN ETHICAL AND EMOTIONALLY INTELLIGENT SOCIAL MEDIA PLATFORM USING RESPONSIBLE AND EXPLAINABLE AI

Project Id:25-26J-137

Project Proposal Report

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Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology

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1.DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Signature of Student		Date	28/8/25
Signature of Supervisor		Date	28/8/25

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2.ABSTRACT

Explainable Friend Recommendation Engine with Interaction Guidance

The growing reliance on social media platforms highlights the need for trustworthy and transparent recommendation systems that can foster meaningful digital interactions. This study focuses on designing an Explainable Friend Recommendation Engine that moves beyond traditional “People You May Know” suggestions by introducing transparency and user-centered engagement. The primary research problem addressed is the lack of clarity in why connections are recommended, which often undermines user trust and can introduce hidden biases. To overcome this, the proposed system will build user interest graphs by analyzing activities such as liked posts, followed topics, and group memberships. A recommendation algorithm based on similarity measures and clustering techniques will be applied to identify potential new connections. To ensure interpretability, the system will incorporate Explainable AI methods such as SHAP and LIME, providing users with clear justifications like “You both follow similar cooking pages” or “You both enjoyed the same movie posts.” Additionally, to enhance interaction quality, the component introduces an Interaction Suggestion Feature that leverages NLP-based templates to generate personalized conversation starters, helping users know how to engage with recommended friends based on shared interests or behavioral cues. This design not only supports fairness by evaluating and minimizing biases in friend recommendations but also strengthens psychological safety and user trust by giving transparency and actionable guidance. By integrating recommendation algorithms with explainable AI and NLP-driven interaction support, this component contributes to building an ethical, emotionally intelligent social media platform where digital connections are both transparent and meaningful.

CONTENTS

1. DECLARATION 3
2. ABSTRACT 4
3. INTRODUCTION 6
3.1. Background: Explainable Friend Recommendation Engine with Interaction Guidance ... 6
3.2. Overall Project Context 6
3.3. Problem Statement Overview 7
4. BACKGROUND AND LITERATURE SURVEY 8
4.1. Background 8
4.2. Literature Survey 8
5. RESEARCH GAP 9
6. RESEARCH PROBLEM 10
7. OBJECTIVES 10
7.1. Main Objective 10
7.2. Specific Objective 10
8. METHODOLOGY 11
8.1. System Architecture (Diagram) 11
8.2. System Solution 12
8.3. Identified Tasks and Sub-tasks 13
8.4. Materials and Resources 13
8.5. Data Requirements and Collection 13
8.6. Work Breakdown Structure 14
8.7. Gantt Chart 15
9. PROJECT REQUIREMENTS 16
9.1. Functional Requirements 16
9.2. Non-Functional Requirements 16
9.3. Expected Test Cases 16
10. REFERENCES 17

3.INTRODUCTION

3.1. Background: Explainable Friend Recommendation Engine with Interaction Guidance

Social media platforms have become central to human interaction, but their recommendation systems often lack transparency. Current "People You May Know" features typically use popularity signals or mutual connections without providing explanations to users. This lack of clarity can reduce trust, reinforce hidden biases, and limit meaningful relationship building. Recent research in **Explainable Recommendation Systems** highlights the importance of making AI-driven suggestions interpretable, so that users understand why a connection is recommended. Techniques such as SHAP and LIME allow developers to generate human-readable explanations for model outputs. At the same time, **Natural Language Processing (NLP)** provides opportunities to go beyond mere explanations by suggesting personalized conversation starters. For instance, if two users share an interest in photography, the system might suggest: "You both enjoy photography — maybe ask about their favorite subject." This engagement-oriented feature introduces an element of emotional intelligence that is largely absent in current systems. Therefore, the Explainable Friend Recommendation Engine with Interaction Guidance aims to enhance trust, fairness, and user engagement by combining interest-based modeling, explainability, and socially supportive interaction prompts.

3.2. Overall Project Context

This component is part of a larger research initiative to design an **Ethical and Emotionally Intelligent Social Media Platform**. The platform integrates four core AI-driven modules:

1. **Cyberbullying Detection & Behavior Profiling** – to detect and manage harmful content in real time.
2. **Emotion-Aware Reaction System & Ethical Search Filtering** – to guide user interactions by respecting emotional tone and blocking harmful queries.
3. **Explainable Friend Recommendation Engine with Interaction Guidance** – to provide transparent, trust-building friend suggestions while helping users engage meaningfully.
4. **Fake Account Detection & User Behavior Analysis** – to identify and restrict suspicious accounts.

Together, these components support fairness, transparency, and digital well-being, addressing critical issues such as safety, trust, and responsible AI in social media environments.

3.3. Problem Statement Overview

The main problem addressed by this component is the **lack of transparency and social usability in friend recommendation systems**. Existing models suggest connections without explaining why, making users question the fairness and accuracy of suggestions. Furthermore, while some explainable recommendation approaches exist, they rarely consider **how users should engage** once a recommendation is made.
This research tackles the problem by:

- Using **explainable AI techniques (SHAP, LIME)** to justify friend recommendations in simple, human-understandable language.
- Applying **NLP template-based generation** to offer personalized, context-aware conversation starters that promote meaningful engagement.
- Evaluating the fairness of recommendations to avoid biases and ensure inclusivity.

By addressing these challenges, the system contributes to creating an ethical, emotionally intelligent platform that improves not only *who* is recommended but also *how* users can form positive social interactions.

4.BACKGROUND AND LITERATURE SURVEY

4.1 Background

Friend recommendation systems are a key feature of social media platforms, designed to increase engagement and connectivity among users. Traditional systems such as Facebook's "People You May Know" rely on features like mutual friends, location, or demographic similarity. While effective in boosting network growth, these systems lack **transparency**, often leaving users unaware of why a recommendation was made. Moreover, hidden biases in recommendation models can reinforce echo chambers and limit diversity in online interactions. As social media increasingly shapes digital identity and relationships, it is important to ensure that friend recommendation systems are both **explainable** and **trustworthy**.

4.2 Literature Survey

Several approaches have been proposed to improve recommendation systems:

- **Collaborative Filtering & Graph-Based Approaches**
  Early research focused on collaborative filtering and graph-based models to recommend users by analyzing network structures and shared connections. Graph-based link prediction methods became widely used, but these approaches lacked personalization beyond social structure.
- **Hybrid and Deep Learning Models**
  More recent systems combine collaborative and content-based features using machine learning or deep learning. For example, Zhang et al. (2019) introduced knowledge-aware and graph neural network-based models that improve accuracy by learning from user interactions and contextual features. While accurate, such models often operate as "black boxes," making their decisions difficult to interpret.
- **Explainable Recommendation Systems (XAI)**
  A growing body of research emphasizes explainability in recommendation. Zhang & Chen (2020) surveyed explainable recommendation techniques, noting that methods such as **SHAP** and **LIME** can highlight the contribution of shared features (e.g., mutual interests) in generating recommendations. These explanations improve user trust and transparency, but most studies stop at providing reasons without supporting the **next step of interaction**.
- **Interaction Guidance in Social Systems**
  Few systems directly support **engagement** after a recommendation. Conversational recommender systems (Gao et al., 2021) explore dialogue-based interactions but are mainly focused on product or item recommendations rather than human-to-human social engagement. This creates a gap for socially intelligent systems that not only explain friend recommendations but also provide **context-aware conversation starters**.

5.RESEARCH GAP

1. **Lack of Transparency in Friend Recommendations**

- Most platforms (Facebook, LinkedIn, Instagram) recommend friends using opaque algorithms based on popularity, mutual friends, or engagement patterns.
- Users are rarely told **why** someone is suggested, which reduces trust (Zhang & Chen, 2020).
- Prior research has introduced explainable recommender systems in movies, e-commerce, and music, but **less in the social networking context**.

2. **Limited Use of Explainable AI in Social Recommendations**

- Some academic works apply SHAP/LIME or graph-based explanations in recommendations (Ribeiro et al., 2016), but few focus on **friend suggestions** where fairness, trust, and transparency are even more critical.
- Current explainability focuses on justification only (eg: "You both liked X") but not on how that explanation can **improve interaction**.

3. **No Support for User Engagement After Recommendation**

- Existing explainable systems stop at explaining *why* a connection is recommended.
- They do not guide users on **how to interact** with the recommended person.
- This limits their usefulness in creating meaningful online relationships.

4. **Ethical & Emotional Intelligence Layer Missing**

- Platforms prioritize growth and engagement rather than **responsible, empathetic interaction**.
- Few systems combine **explainability, fairness, and emotional intelligence** in one design.

6. RESEARCH PROBLEM

Most social media platforms recommend new connections through opaque algorithms based on popularity, mutual friends, or activity patterns. While these recommendations increase user engagement, they suffer from two major limitations. First, they lack **transparency**: users are not informed why a particular connection is suggested, which reduces trust and raises concerns of algorithmic bias (Zhang & Chen, 2020). Second, they do not support **meaningful interaction** once a recommendation is made. Current explainable recommender systems in e-commerce and entertainment domains provide justifications such as "You liked similar items," but in the **social networking context**, friend recommendation systems rarely guide users on how to engage with new connections.

This lack of transparency and engagement support creates a **research gap**: the absence of explainable and socially supportive friend recommendation engines. Addressing this gap requires not only generating interpretable explanations (using Explainable AI) but also extending recommendations with **NLP-driven interaction suggestions**, enabling users to start conversations confidently and meaningfully.

7. OBJECTIVES

7.1 Main Objective

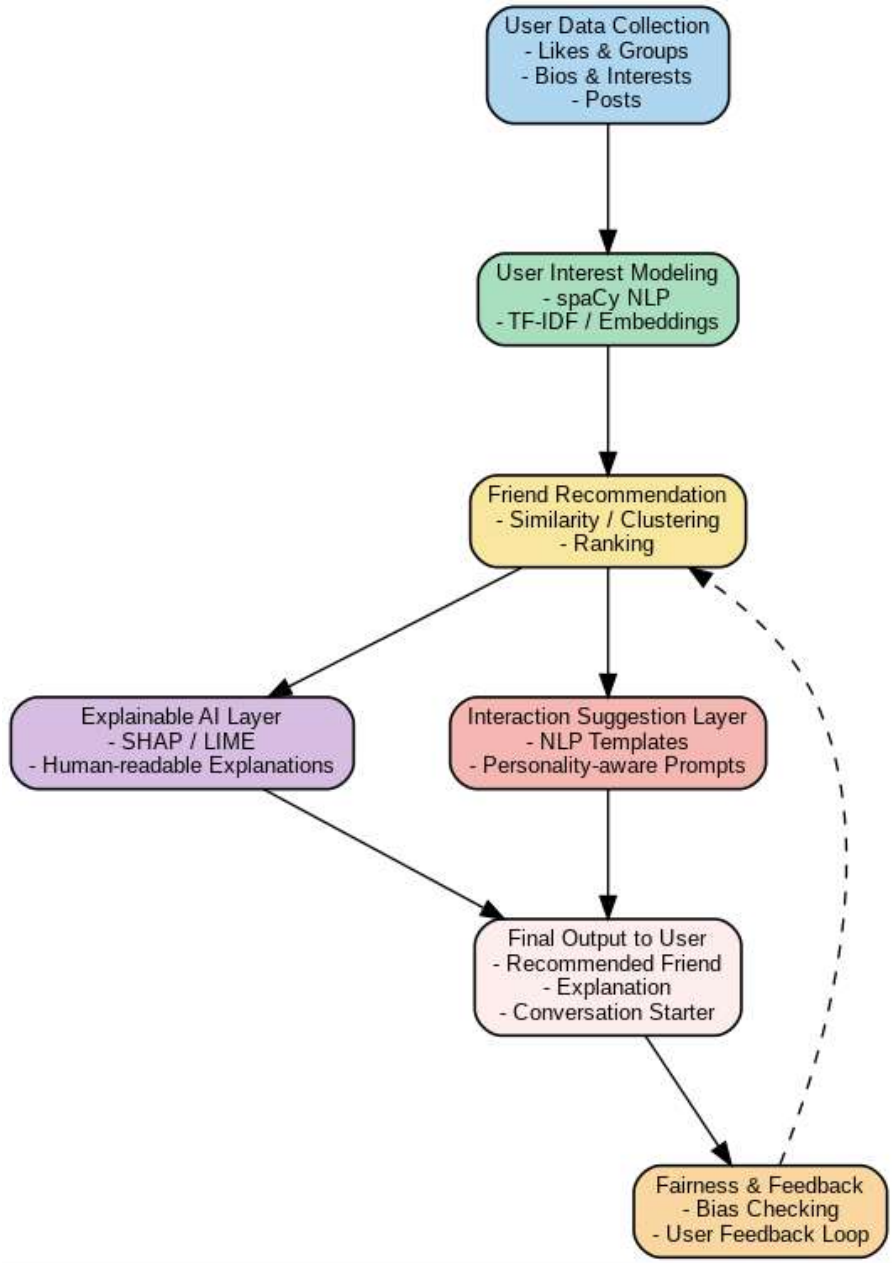
To design and develop an **Explainable Friend Recommendation Engine with Interaction Guidance** that recommends new connections based on shared interests, explains the reasoning behind suggestions using Explainable AI, and provides context-aware conversation starters to help users engage meaningfully.

7.2 Specific Objectives

- To model user interests by analyzing activity data such as liked posts, followed topics, and group memberships.
- To apply similarity-based or clustering algorithms for identifying potential new connections.
- To integrate Explainable AI techniques (e.g., SHAP, LIME) to provide transparent reasons for each recommendation.
- To develop an **NLP template-based system** that generates personalized conversation starters based on shared interests.
- To evaluate fairness and reduce algorithmic biases in recommendations.
- To incorporate user feedback for refining the accuracy, fairness, and usefulness of recommendations.

8. METHODOLOGY

8.1 System Architecture Diagram



8.2 System Solution

The proposed solution is an **Explainable Friend Recommendation Engine with Interaction Guidance** designed to make friend suggestions transparent, fair, and socially supportive. The system follows a modular architecture with four main layers:

1. **User Interest Modeling**
   - Collect user activity data (liked posts, followed groups, profile keywords, bios).
   - Preprocess text using **spaCy** to extract interest keywords and build user interest profiles.
   - Represent users as vectors (TF-IDF or embeddings) for similarity analysis.
2. **Friend Recommendation Module**
   - Apply **similarity-based or clustering algorithms** (e.g., cosine similarity, K-means) to identify potential new connections.
   - Rank and filter recommendations based on shared interests.
3. **Explainable AI Layer**
   - Integrate **SHAP/LIME** to explain why a recommendation is made.
   - Explanations are converted into human-readable sentences (e.g., "You both liked cooking and travel-related posts").
   - Optional: Multiple explanation styles (simple, detailed, visual) to improve user control and trust.
4. **Interaction Guidance Module**
   - Implement **NLP template-based generation** to provide personalized conversation starters.
   - Example: If two users share "photography," the system suggests: *"You both enjoy photography — ask what camera they use."*
   - Extend with **personality-aware prompts** using lightweight personality cues (e.g., emojis, bio tone) for better alignment.
5. **Fairness and Feedback Mechanism**
   - Introduce a **fairness audit check** to ensure recommendations are balanced and not biased toward only popular or highly active users.
   - Allow users to give feedback ("Was this suggestion helpful?") to refine both explanations and interaction templates over time.

This system solution addresses the research problem by **making recommendations explainable, socially meaningful, and ethically fair**, ultimately improving both trust and engagement on the platform.

8.3 Identified Tasks and Sub-tasks

1. **Data Collection & Preprocessing**
   - Gather user activity data (likes, topics, groups).
   - Clean and preprocess text data.
   - Extract interest vectors.
2. **User Interest Modeling**
   - Build user profiles with embeddings or TF-IDF vectors.
   - Cluster or calculate similarity scores.
3. **Recommendation Algorithm**
   - Apply cosine similarity / clustering for friend suggestions.
   - Rank potential friends.
4. **Explainability Layer**
   - Apply SHAP/LIME to justify recommendations.
   - Display reasons in human-readable form (e.g., “You both liked cooking posts”).
5. **Interaction Suggestion Feature**
   - Use NLP templates to generate conversation starters.
   - Enhance with spaCy keyword extraction.
6. **Evaluation**
   - Test for fairness, accuracy, and user satisfaction.
   - Collect user feedback.

8.4 Materials and Resources

- **Software/Frameworks**: Python, scikit-learn, pandas, NumPy, spaCy, SHAP, LIME.
- **Backend**: Flask / FastAPI.
- **Frontend**: React.
- **Hardware**: Standard PC/laptop (8GB+ RAM recommended).
- **Data Storage**: MySQL or PostgreSQL.

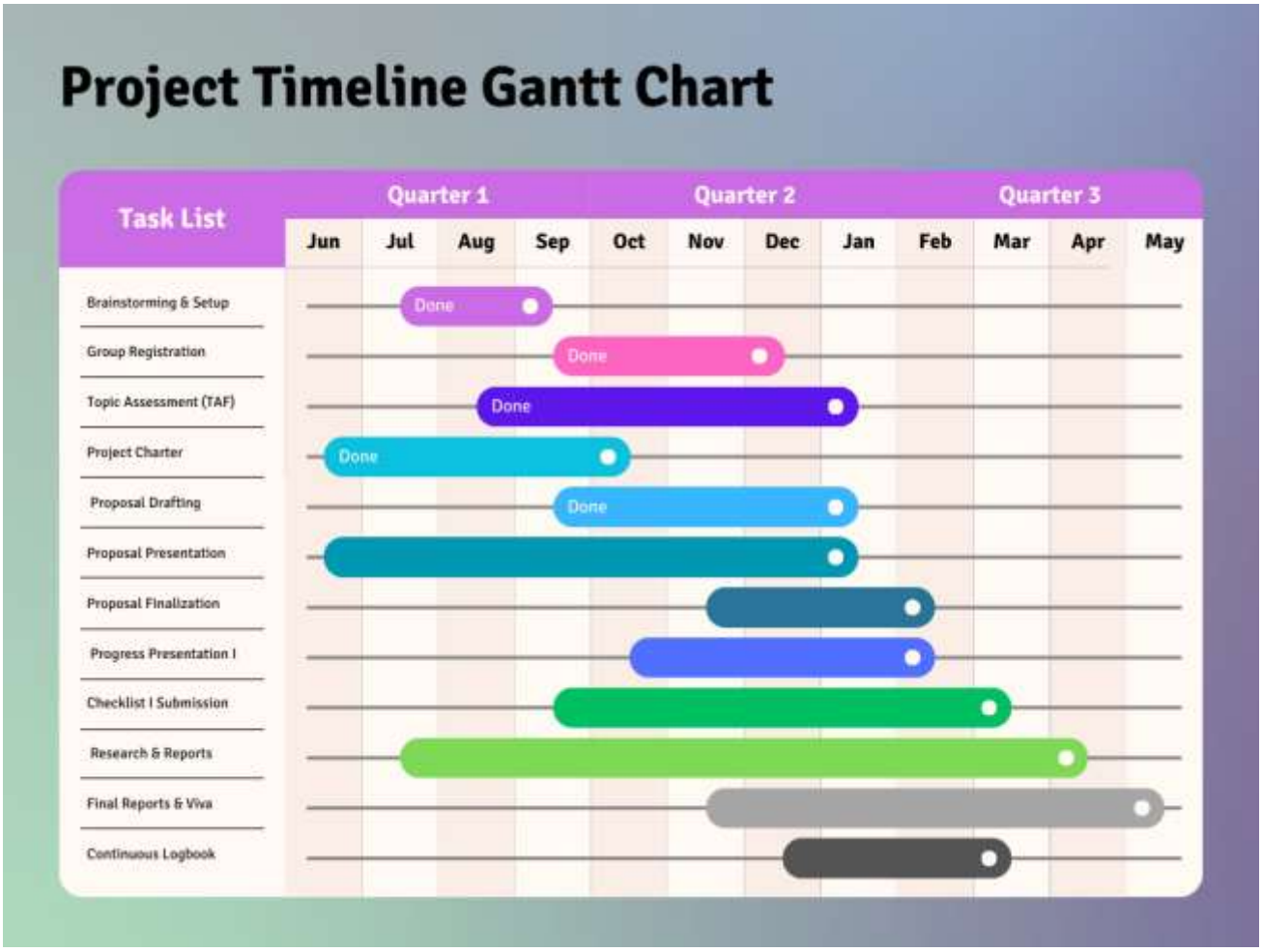
8.5 Data Requirements and Collection

- User Interaction Data: Posts liked, topics followed, groups joined.
- Behavioral Metadata: Frequency of interactions, engagement levels.
- User Bios/Profiles: Text data for keyword extraction.
- Sample Datasets:Publicly available **Twitter dataset** (GitHub: *Social Media Dataset Samples*).

8.6 Work Breakdown Structure

Task	Subtasks
Data Preparation	Collection, cleaning, preprocessing
Interest Modeling	Feature engineering, similarity graph
Recommendation Algorithm	Similarity matching, ranking
Explainability	SHAP/LIME integration, justification formatting
Interaction Guidance	NLP templates, spaCy keyword extraction
Evaluation	Fairness testing, user study, feedback loop
Integration	Backend API, frontend interface

8.7 Gantt Chart



9. PROJECT REQUIREMENTS

9.1 Functional Requirements

- System must generate friend recommendations based on user interests.
- Each recommendation must be accompanied by an explanation (via XAI).
- System must provide at least one conversation starter per recommendation.
- Users must be able to give feedback on recommendations.

9.2 Non-Functional Requirements

- **Transparency**: Explanations should be human-readable.
- **Fairness**: Avoid bias (e.g., recommending only highly active/popular users).
- **Performance**: Generate recommendations in real time (<2 seconds).
- **Scalability**: Support increasing numbers of users.
- **Usability**: Interface must be intuitive and user-friendly.

9.3 Expected Test Cases

- Test whether recommendations change based on shared interests.
- Test explanation accuracy (does SHAP/LIME highlight the correct reason?).
- Test if interaction guidance is context-aware (cooking → recipe starter).
- Test fairness (e.g., diverse recommendations).
- Test user satisfaction through surveys.

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